



产品定作合同

甲方（定作方）：苏州节奏能源科技有限公司

乙方（加工方）：江苏瑞恩电气有限公司

合同编号：RE3231201-01

项目名称：美国美变项目

为保证双方的合法利益，本着平等协商、互利互惠的原则，双方依据《民法典》以及相应法规，一致同意按下列条款签订本合同。

一、产品名称、规格、型号、数量及金额

签订日期：2023年12月1日

产品名称	规格型号	描述	单位	数量	单价 (元)	合计 (元)	备注
变压器	ZGSL-H-1500/ 12.47Y/7.2 /0.48Y/0.277 kv	60Hz,YNyn0, Loop feed, color: telephone green(Munsell No.7.0 GY3.2/1.5)	个	1	150000	150000	调压：±2x2.5% FR3 油 硅钢铁芯 全铝，阻抗 6.3%， 低压套管 12 个 孔， 空载损耗：1300W 负载损耗： 14000W 结构：三相三柱

注：此价格含至上海港运费、含税(13%增值税发票)；

1. 产品标识、外包装须满足出口要求且做好防潮湿处理，
2. 变压器铭牌按甲方提供的铭牌模板制作，
3. 变压器必须在确认图纸后才能进行生产，
4. 详细技术要求见下面附件。

合计人民币总金额（大写）：壹拾伍万元整

(¥:150000.00 元)

二、质量标准：按照合同后附技术协议和图纸生产，质量保证期为发货之日起 18 个月，不含国外售后服务费。

三、交货期限：合同签订生效后即可安排生产，35 天具备发货条件（遇突发事件或者不可抗拒因素除外），甲方按时提货。

四、运输及交货：在甲方厂内交货或者上海，合格后出厂。

五、产品验收：甲方组织工厂验收，如对验收有异议可在发货前提出，否则视为验收合格。

六、付款方式：1、预付 20%，发货前付清全款。

2、甲方应将货款汇入乙方指定账户，现金结算，不得以承兑结算。

七、本合同中产品所有权自货款付清时起转移，货款未付清的，该产品所有权仍属于乙方所有。

八、本合同生效方式及违约责任：双方盖章即生效。如逾期付款每逾期一日应按照逾期金额的万分之五标准计算违约金，同时还应赔偿由此造成的全部经济损失（包括但不限于诉讼费、保全费、律师费、因保全而产生的保险费等）。若发生争议，依法向原告方所在地法院诉讼。

九、其他约定事项：

1、甲方指定的收货人：_____，联系电话：_____，所在单位：_____。

2、每季度第 1 个月 1-15 日为乙方财务对帐日，甲方必须配合乙方财务做好往来核对。

3、本合同下的产品及包装专为甲方定制，合同生效后因任何一方的过错导致合同的解除或变更视为违约，违约方需向守约方支付合同总额 20% 的违约金，如合同签订日后 3 个月内甲方未提货，则乙方有权单方解除合同，甲方应承担违约责任及赔偿全部损失（包括但不限于诉讼费、保全费、律师费、因保全而产生的保险费等）。

4、本合同单方涂改无效，双方盖章后的传真件、电子邮件、微信图片等与原件具有同等法律效力。

5、如合同签订日当天，双方未盖章回传或预付款未到乙方账户，则本合同下定作产品价格须根据原材料价格涨幅上调；如本合同无预付款，合同签订日后 45 日内甲方未提货，则本合同价格及交货期须重新调整。



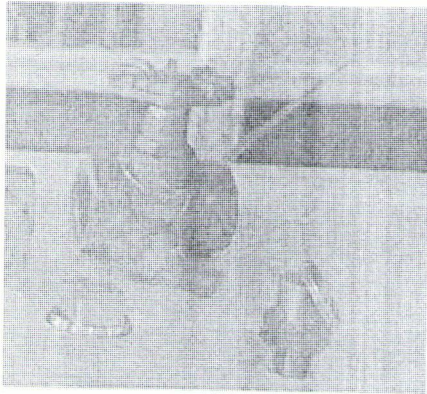
甲方：
单位名称（章）：
单位地址：
法定代表人：
委托代理人：
电 话：
传 真：
开户银行：
税 号：
账 号：



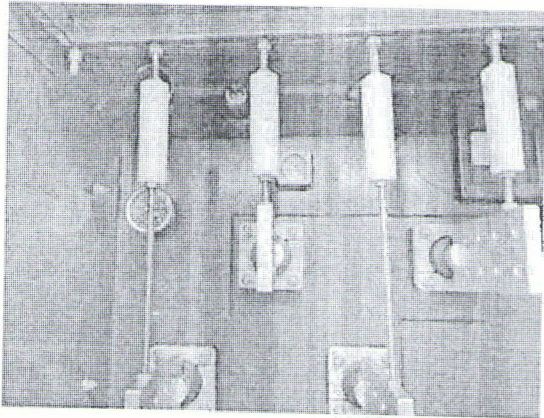
乙方：
单位名称（章）：江苏瑞恩电气有限公司
单位地址：江苏·海安黄海大道西 229 号
法定代表人：王良明
委托代理人：
电 话：0513-88699099
传 真：0513-88837088
开户银行：中国农业银行股份有限公司海安市支行
税 号：91320600662127205A
账 号：10700101040219916

技术要求附件：

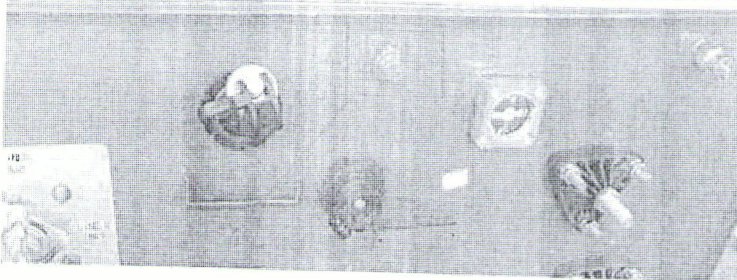
- 1, 内部连接导线必须用多股软铜线加铜鼻子接头。
- 2, 采用此款放油阀，位于低压室内。



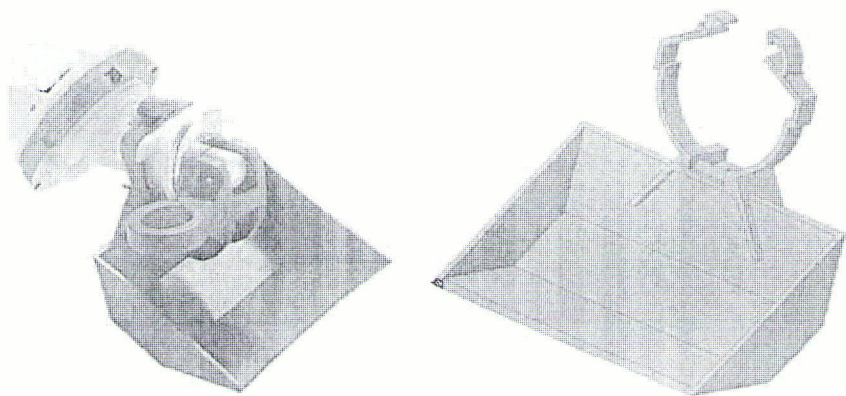
- 3, 低压套管要安装撑杆，如图：



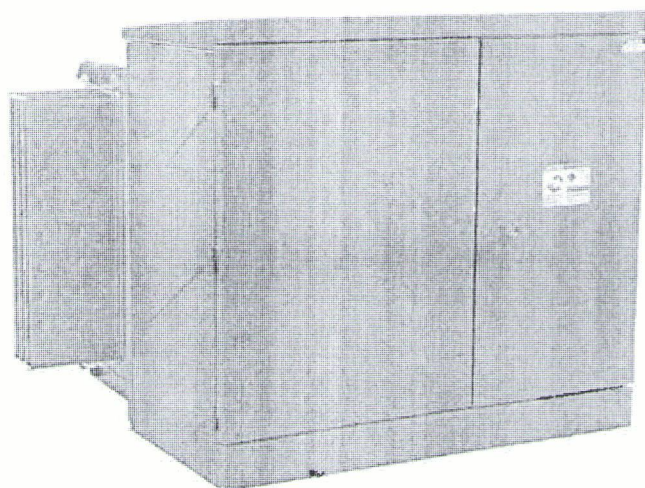
- 5, 注油口采用直角设计，不要倾斜，采用单相美变的设计，如图：



- 6, 熔丝套管套一个集油盒，如图：

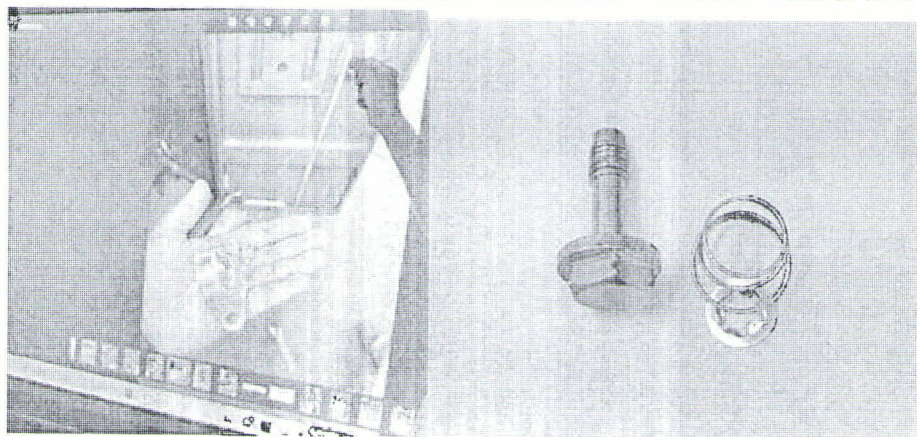


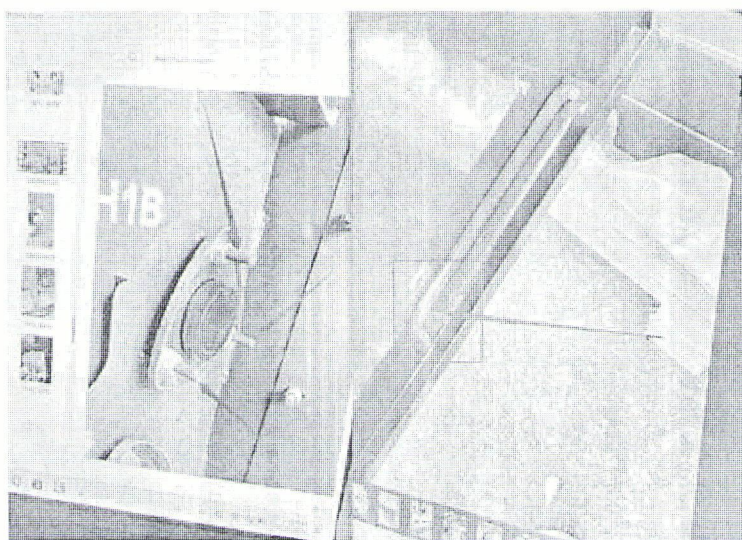
7, 高低压室尺寸要完全按照美标做, 前舱采用此款设计, 门栓在侧面, 这样门可以开成 180 度, 如图:



8, 变压器内外都不需要任何印字, 表阅标签也不要印, 也不要门上装危险提示牌, 两个铭牌, 一个在低压室, 一个在低压室门内测。

9, 门把手螺栓材质需是 304 不锈钢的, 并且要加一个弹簧, 高压套管上螺丝不能生锈, 等, 如图:





10, 油箱变压器室上的盖子要固定住。

11, 变压器油箱和前仓需安装固定板, 以便变压器可以固定安装在水泥基座上。

技术要求:

- KNAN, FR3 植物油, non-PCB

4. Each distribution transformer shall be equipped with a pressure relief valve.

5. Transformer taps (when specified) shall be 2 - 2 1/2% below and 2 - 2 1/2% above rated voltage. The operating mechanism of the tap changer is to be located in the high voltage compartment. It shall be clearly marked and have a mechanism to secure the switch in its current position to prevent accidental movement. The switch shall be secured in the 100% position at the time of shipment.

6. All transformers will be designed for loop feed with six (6) 25kV 200 amp high voltage bushing wells mounted as per ANSI Standard C57.12.26 - Figure 6 (latest revision). They shall be of the type that is clamped or bolted to the faceplate. This interface is to meet ANSI Standard 386 (latest revision). All bushing wells shall have protective covers attached to prevent contamination during shipment and storage. To meet the dimensional requirements of Table 1, a K=5.0 is acceptable as listed in ANSI Standard C57.12.26 - Figure 6a. Similarly, to meet the dimensional requirements of Table 1, an F=18 is acceptable as listed in ANSI Standard C57.12.26

- Figure 7.

7. Secondary terminals to be fully insulated NEMA drilled spades. They shall be of the type that is clamped or bolted to the faceplate. Neutral terminal to be fully insulated NEMA drilled spade. Each spade shall contain no less than six (6) holes and conform to ANSI Standard C57.12.34 (latest revision) Figure 9b ONLY. The positioning of the secondary terminals shall conform to ANSI Standard C57.12.34 (latest revision) Figure 8a.

8. A ground pad and a removable copper ground strap shall be provided and connected to the neutral spade in such a way as not to subtract from the number of spade holes available.

9. Three (3) 'Bay-o-net' type fuse holders and drip pans are required on 75 - 1500 kVA units. Fuse sizes and types as per Table 1.

10. Three (3) under oil current limiting backup fuses are required on 75-1500 kVA units. Fuse sizes and types as per Table 1.

11. A minimum of five (5) parking stands is required.

12. A minimum of six (6) transformer tank grounding connectors required and located as per figure 2. These shall be Blackburn TTC2 (or equivalent).

13. The transformer rating is based on a 65°C temperature rise.
14. A NEMA "DANGER - HIGH VOLTAGE" sign shall be placed on the inside of the transformer per NEMA Standard 260, Figure 2 (latest revision).
15. Oil fill and oil drain plugs shall be located in the low voltage compartment for 75 – 750 kVA transformers. Transformers 1,000 kVA and larger shall have the oil fill plug located in the low voltage compartment and an external drain valve with a sampling device. The location of the drain valve and sampler shall be placed externally on the low voltage tank side and shall be provisioned to be concealed and protected from entry by a lockable box equipped with a recessed Penta-head bolt. The box shall be sized to sufficiently cover the valve and removable only after the lock has been removed and the bolt loosened.
16. The liquid level gauge shall be located in the low voltage compartment of 1,000 kVA and larger transformers.
17. The transformer enclosure shall meet the Western Underground Committee Guide No. 2.13 (latest revision).
18. The locking device on the low voltage compartment door is to consist of a captive stainless steel Penta-head bolt (1/2 - 13 UNC) in a non-rotating cup. It must have provision for a padlock arranged in such a way that the lock must be removed prior to any access to the Penta-head bolt. In addition, a second fastening device must be attached to the high voltage compartment door. It must be arranged in such a way that the low voltage compartment door must be opened first to allow access to the high voltage compartment fastening device.
19. The exterior color of all transformers will be telephone green (Munsell No. 7.0.GY3.2/1.5). The minimum dry paint film thickness shall be 3 mils.
20. The bottom 2" of the tank shall be treated to minimize corrosion.
21. Four (4) lifting lugs are required.
22. Transformers, 750 kVA and larger, shall have a Celsius temperature gauge located in the low voltage compartment.
23. All transformers will be subtractive polarity.
24. Transformer impedance shall be in accordance with the table below:

Table 1 Transformer Impedance

kVA Rating	Min. Impedance in %	Max. Impedance in %
75	2.0	3.5
150	2.0	4.5
225	3.5	4.5
300	3.5	4.5
500	4.0	6.0
750	5.3	6.3
1,000	5.3	6.3
1,500	5.3	6.3
2,000	5.3	6.3
2,500	5.3	6.3

